Jacob “Lyla” Noe

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Software engineer with 4+ years of experience building and maintaining production-grade web and mobile systems. Promoted to lead developer in 2025, with end-to-end ownership across Python/Flask, TypeScript/React, PostgreSQL, Redis, Docker, AWS, and CI/CD. Independent work in associative memory retrieval, longitudinal behavioral modeling, and sensor-driven interfaces, with an emphasis on explicit architecture, automated testing, reproducibility, and honest evaluation.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C, C++, Kotlin, Swift

Frameworks and Data: Flask, FastAPI, React, React Native / Expo, PostgreSQL, MySQL, Redis, D3.js, Socket.IO

Testing and Delivery: Git, GitHub, GitHub Actions, AWS CodeBuild, Jest, Cypress, pytest, Docker, AWS ECS and Lambda, CI/CD

Architecture: Offline-first systems, real-time synchronization, event-driven automation, distributed resource coordination

Current Technical Study: PyTorch model development and reproducible training workflows; Kubernetes container orchestration fundamentals

PROFESSIONAL EXPERIENCE

Hero Services | Subsidiary of Ellison Environmental *2022-Present*

Lead Full Stack Developer, 2025-Present | Full Stack Developer, 2022-2025

- Own a production platform supporting water and wastewater facility operations across multiple client organizations, including web and mobile applications, backend services, deployments, and production support.
- Designed offline-first React Native / Expo workflows with durable local storage, network-aware request queuing, and synchronization for unreliable field connectivity.
- Implemented real-time synchronization between React clients and Flask using Socket.IO, with Redis-backed room membership and resource locks for concurrent users.
- Built checksheet-driven dynamic forms, custom D3 visualizations, reporting systems, data-processing features, and scheduled or event-driven notifications.
- Maintained pull-request-triggered GitHub Actions workflows that dispatched AWS CodeBuild pipelines, running Jest and Cypress for client changes and pytest for API changes targeting staging and master branches.
- Maintained component-specific build and deployment workflows for React, Flask, and Lambda services; operated Docker-based frontend and API deployments through AWS ECS.
- Improved application security through a password-hashing migration and production authentication updates.
- Documented complex logic through targeted source comments and explained code changes, architecture, and logic flows to teammates to support maintainability and knowledge transfer.

California Polytechnic State University *2020-2021*

Full Stack Developer | Dr. Brennan Davis

- Served as sole developer, leading the design and implementation of public-facing websites for university faculty.
- Developed interfaces, database architecture, and API integrations using JavaScript, React, Python, PHP, and MySQL; managed domain configuration.

iOS App Developer | Dr. Brennan Davis and Dr. Suzanne Phelan | 2020

- Developed an augmented-reality experience and a diet-tracking application using Swift and Unity / C#.
- Integrated barcode scanning, nutrition APIs, and 3D coordinate systems into mobile user experiences.

SELECTED PROJECTS

Orbit | *Longitudinal behavioral modeling*

- Built a functioning Android application for logging mood, sleep, activity, diet, exercise, medication, and biometric inputs using encrypted local storage.
- Implemented a ten-axis state encoder, approximate nearest-neighbor indexing, and density-based grouping over longitudinal observations; added synthetic-data generation and unit tests while keeping association explicitly separate from causation.

SELECTED PROJECTS

Associative Memory Harness | ***Cognitive-inspired retrieval***

- Built a Python retrieval system combining semantic embeddings with bounded recursive traversal of a weighted memory graph whose links strengthen when memories are retrieved together.
- Added a separate emotional-association channel, persistent graph state, injectable deterministic embeddings, and regression tests; identified baseline evaluation as the next research requirement.

Whorl | ***Sensor-driven musical interface***

- Integrated an Arduino GIGA R1, Hall sensor, and magnet-equipped spinner into a working controller for synthesizer modulation in Logic / Alchemy.
- Developed C++ firmware for pulse processing and RPM estimation plus a Python serial-to-MIDI bridge; documented the protocol and tested the production firmware against simulated sensor input.

ADDITIONAL EXPERIENCE

From War to Peace *2025*

Shopify Developer

- Worked within an existing Shopify website, integrating applications and implementing custom solutions in Liquid.

Early-Stage Startup *2021*

Co-Founder and Full Stack Developer | Homelessness Solutions

- Co-founded a city-funded startup focused on improving access to employment pathways for unhoused individuals.
- Designed and built a D3 node-graph interface for exploring employment opportunities.

EDUCATION

California Polytechnic State University *December 2021*

Bachelor of Science in Computer Science | Cum Laude | GPA 3.56